

- (i) State an expression, in terms of R and ω , for the speed v of the ball.

$$v = \dots\dots\dots [1]$$

- (ii) Determine an expression, in terms of v and ω , for the centripetal acceleration of the ball.

$$\text{centripetal acceleration} = \dots\dots\dots [2]$$

- (c) The ball in (b) is in the position shown in Fig. 1.1, such that line OB is at an angle θ to the line OP.

- (i) Determine an expression, in terms of R and θ , for the displacement x of the shadow from P.

$$x = \dots\dots\dots [1]$$

- (ii) The value of θ is zero at time $t = 0$.

State an expression for θ in terms of ω and t .

$$\theta = \dots\dots\dots [1]$$

- (iii) Use your answers in (c)(i) and (c)(ii) to show that x is given by

$$x = R \sin \omega t.$$

[1]

- (iv) Explain, with reference to the equation in (c)(iii), why the motion of the shadow of the ball on the screen may be modelled as simple harmonic.

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 [1]

